

Exact Evaluation of the Jones Matrix of a Fiber in the Presence of Polarization Mode Dispersion of Any Order

Enrico Forestieri, *Member, IEEE*, and Luca Vincetti

Abstract—This paper describes how to evaluate exactly in a recursive manner the Jones transfer matrix of a fiber for any order of polarization mode dispersion (PMD) and derives for it a second-order accurate analytical approximation. The second-order approximation is then compared against the exact result as well as with other models found in the literature.

Index Terms—Optical fiber communication, optical fiber polarization, polarization mode dispersion (PMD).

I. INTRODUCTION

WITH the advent of long-distance high bit rate optical systems, polarization mode dispersion (PMD) has become an important source of limitation for the system performance. PMD causes a broadening of the transmitted optical pulse, which depends on the input field polarization and exhibits a stochastic behavior. PMD may be usefully described by means of the fiber Jones matrix [1], which relates the input and output electric fields. Alternatively, PMD may be characterized in the Stokes space by means of the dispersion vector [2]. The analysis of the PMD impact on the system performance is easily worked out by means of the Jones matrix, but unfortunately, until now, the statistical characterization of such matrix has not been known. On the contrary, the joint statistics of the dispersion vector and its frequency derivative [3]–[5] are known, but, working in the Stokes space, only the broadening factor of the optical signal can be evaluated [6], [7]. As a consequence, if the goal is to develop a useful model for an accurate analysis of the system performance, the knowledge of the analytical relationship between the Jones matrix and the dispersion vector is mandatory.

In the first-order approximation, the dispersion vector is assumed to be independent from frequency over the signal bandwidth. In this case, the corresponding Jones matrix is known and the output pulse is composed by two replicas of the transmitted one, differentially delayed by a quantity equal to the modulus of the dispersion vector. For wide-band signals, the frequency dependence of the dispersion vector generates higher order effects, in particular, second-order effects. In this latter case, the dispersion vector may be approximated by its Taylor expansion truncated to the first order [3].

In recent years, several approximate expressions of the Jones matrix for the analysis of the second-order PMD effects have been developed [8]–[12]. Bruyère, in [9], proposed an extension of the first-order matrix supposing that, in the Stokes space, the directions of the Jones matrix eigenvectors and the dispersion vector coincide. Strictly speaking, this is correct only for the first-order approximation; thus, this model overestimates the second-order effects by a factor of two [10], [11]. To overcome this problem, in [10], Kogelnik *et al.* demonstrated how to improve the accuracy of the model, whereas in [11], a new solution, which is correct when the input dispersion vector rotates on the equatorial plane maintaining the same modulus, was proposed. Recently, a new description of the Jones matrix based on an exponential expansion has been presented [12]. This approach arbitrarily assumes that the matrix factors into the product of several matrices, one for each order of PMD, which are then determined by equating the successive derivatives with respect to frequency.

In this work, an exact power series expansion of the Jones matrix is presented, allowing the description of the behavior of the fiber and the evaluation of its performance [13] for any given amount of PMD of any order. Our interest concentrates then on the second-order PMD, as statistics for this order are known, for which a very accurate approximate analytical expression is also presented. The Jones matrix is directly related to first- and second-order dispersion vector parameters; thus, the stochastic PMD behavior can be taken into account immediately. Comparison with other second-order models shows that our analytical Jones matrix approximate expression is more accurate. The paper is organized as follows. In Section II, the basic PMD theory is reported and used to obtain the differential equation that relates the Jones matrix and the dispersion vector. The exact power-series expansion of the differential equation solution and its analytical approximation are given in Section III. In Section IV, a comparison between the exact solution and the new approximation, as well as other approximations proposed in literature, is reported.

II. PMD THEORY

The Fourier transform of the low-pass equivalent of a modulated optical signal at the laser output may be written as

$$\hat{S}_{\text{in}}(\omega) = S(\omega)\hat{e}_{\text{in}} \quad (1)$$

where $S(\omega)$ is a scalar quantity and $\hat{e}_{\text{in}}$ is a unitary vector (with complex components) representing the electric field

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E. Forestieri is with the Scuola Superiore Sant'Anna di Studi Universitari e Perfezionamento, Pisa, Italy.

L. Vincetti is with the Consorzio Nazionale Interuniversitario per le Telecomunicazioni (CNIT), Pisa, Italy.

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polarization state and is generally known as the polarization Jones vector. As the output laser light is completely polarized, all signal frequency components have the same polarization state and $\hat{e}_{\text{in}}$ is frequency independent. When referred to a right-handed coordinate system (x, y, z) oriented such that z coincides with the propagation direction and the electric field is transversal to this direction, $\hat{e}_{\text{in}}$ is a bidimensional vector representing the electric field lying in the (x, y) plane.

In the linear regime, a birefringent lossless monomodal fiber of length L is characterized by the Jones matrix

$$T(\omega) = e^{-j\beta(\omega)L}U(\omega) \quad (2)$$

where $\beta(\omega)$ is the propagation constant accounting for the fiber chromatic dispersion, and

$$U(\omega) = \begin{pmatrix} u_1(\omega) & u_2(\omega) \\ -u_2^*(\omega) & u_1^*(\omega) \end{pmatrix} \quad (3)$$

is a unitary matrix, i.e., $|u_1(\omega)|^2 + |u_2(\omega)|^2 = 1$, accounting for PMD.

At the fiber output, in the frequency domain, the signal may be written as

$$\begin{aligned} \hat{S}_{\text{out}}(\omega) &= T(\omega)S(\omega)\hat{e}_{\text{in}} \\ &= e^{-j\beta(\omega)L}U(\omega)S(\omega)\hat{e}_{\text{in}} \\ &= S_o(\omega)\hat{e}_o(\omega) \end{aligned} \quad (4)$$

where

$$S_o(\omega) \triangleq S(\omega)e^{-j\beta(\omega)L} \quad (5)$$

and

$$\hat{e}_o(\omega) \triangleq U(\omega)\hat{e}_{\text{in}} \quad (6)$$

is the output polarization Jones vector. In the Stokes space, (6) takes the form

$$\vec{e}_o(\omega) = M(\omega)\vec{e}_{\text{in}} \quad (7)$$

where $M(\omega)$ is a 3×3 real unitary matrix known as the fiber Müller matrix, and $\vec{e}_{\text{in}}$ and $\vec{e}_o(\omega)$ are real three dimensional unitary vectors corresponding to $\hat{e}_{\text{in}}$ and $\hat{e}_o(\omega)$, respectively. We adopt the convention of using a hat for vectors in the Jones space and an arrow for vectors in the Stokes space.

In order to relate to each other the matrices in the Stokes and Jones spaces so that we can evaluate $U(\omega)$ when we know $M(\omega)$, we reason in the following way. The matrix $M(\omega)$ may be interpreted geometrically as a rigid rotation of an angle $2a(\omega)$ about the unit vector $\vec{b}(\omega) = [b_1(\omega), b_2(\omega), b_3(\omega)]^T$ and can be expressed as [6]

$$\begin{aligned} M(\omega) &= \exp[2a(\omega)\vec{b}(\omega) \times] \\ &= \cos 2a(\omega)I + [1 - \cos 2a(\omega)]\vec{b}(\omega)\vec{b}(\omega)^T \\ &\quad + \sin 2a(\omega)\vec{b}(\omega) \times \end{aligned} \quad (8)$$

where I is the identity matrix, $\vec{b}(\omega)\vec{b}(\omega)^T$ is a dyad, and $\vec{b}(\omega) \times$ is the skew-symmetric matrix corresponding to the cross-product operator

$$\vec{b}(\omega) \times = \begin{pmatrix} 0 & -b_3(\omega) & b_2(\omega) \\ b_3(\omega) & 0 & -b_1(\omega) \\ -b_2(\omega) & b_1(\omega) & 0 \end{pmatrix}. \quad (9)$$

As M represents a rotation, $\det[M] = 1$, and from (8), trace $[M] = 1 + 2\cos 2a(\omega)$. Denoting by λ_i , $i = 1, 2, 3$, the eigenvalues of M ($|\lambda_i| = 1$ as M is unitary), since the same quantities may also be written as $\det[M] = \lambda_1\lambda_2\lambda_3$ and $\text{trace}[M] = \lambda_1 + \lambda_2 + \lambda_3$, it follows that $\lambda_1 = 1$, $\lambda_2 = e^{j2a(\omega)}$ and $\lambda_3 = e^{-j2a(\omega)}$. It may be shown that $\vec{b}(\omega)$ is the normalized eigenvector corresponding to λ_1 and that, assuming $b_1 \neq \pm 1$ and $b_3 \neq 0$

$$\begin{aligned} \cos 2a &= \frac{m_{11} - b_1^2}{1 - b_1^2} \\ \sin 2a &= \frac{m_{21}}{b_3} - \frac{b_1 b_2}{b_3} \frac{1 - m_{11}}{1 - b_1^2} \end{aligned} \quad (10)$$

where $m_{ij}(\omega)$ are the elements of the matrix $M(\omega)$.

Thus, from $M(\omega)$, we can determine $\vec{b}(\omega)$, $\cos a(\omega)$ and $\sin a(\omega)$, and, hence, the Jones matrix $U(\omega)$, as it may be expressed as [6]

$$\begin{aligned} U(\omega) &= \exp[ja(\omega)B(\omega)] \\ &= \cos a(\omega)I + j \sin a(\omega)B(\omega) \end{aligned} \quad (11)$$

where $B(\omega)$ is a Hermitian 2×2 matrix related to the components of the unit vector $\vec{b}(\omega)$ according to

$$B(\omega) = \begin{pmatrix} b_1(\omega) & b_2(\omega) - jb_3(\omega) \\ b_2(\omega) + jb_3(\omega) & -b_1(\omega) \end{pmatrix}. \quad (12)$$

For fixed-input polarization $\vec{e}_{\text{in}}$, PMD causes the output polarization $\vec{e}_o(\omega)$ to spin around the Poincaré sphere according to [14]

$$\begin{aligned} \frac{d\vec{e}_o(\omega)}{d\omega} &= \vec{\Omega}(\omega) \times \vec{e}_o(\omega) \\ &= \Phi(\omega)\vec{e}_o(\omega) \end{aligned} \quad (13)$$

where $\vec{\Omega}(\omega) = [\Omega_1(\omega), \Omega_2(\omega), \Omega_3(\omega)]^T$ is the PMD vector and $\Phi(\omega) \equiv \vec{\Omega}(\omega) \times$ is the 3×3 matrix

$$\Phi(\omega) = \begin{pmatrix} 0 & -\Omega_3(\omega) & \Omega_2(\omega) \\ \Omega_3(\omega) & 0 & -\Omega_1(\omega) \\ -\Omega_2(\omega) & \Omega_1(\omega) & 0 \end{pmatrix}. \quad (14)$$

By substituting (7) in (13), we find that $M(\omega)$ is related to $\vec{\Omega}(\omega)$ through

$$\frac{dM(\omega)}{d\omega} = \Phi(\omega)M(\omega) \quad (15)$$

which we could use to determine $M(\omega)$ and then $U(\omega)$, as explained previously.

Alternatively, it is simpler to directly relate $U(\omega)$ to $\vec{\Omega}(\omega)$ in the following way.

It is known that the unit vectors representing the polarization states parallel and antiparallel to $\vec{\Omega}(\omega)$, often referred to as the principal states of polarization (PSP) [15], are, in the Jones space, the normalized eigenvectors of the matrix

$$Q(\omega) = -j \frac{dU(\omega)}{d\omega} U^{-1}(\omega) \quad (16)$$

which can be easily shown to be Hermitian of the form

$$Q(\omega) = \begin{pmatrix} q_1(\omega) & q_2^*(\omega) \\ q_2(\omega) & -q_1(\omega) \end{pmatrix} \quad (17)$$

where $q_1(\omega)$ is real. A simple calculation shows that the eigenvalues of $Q(\omega)$ are

$$\lambda_{\pm}(\omega) = \pm \sqrt{q_1^2(\omega) + |q_2(\omega)|^2} \quad (18)$$

and are related to $\vec{\Omega}(\omega)$ through

$$|\lambda_{\pm}(\omega)| = \frac{1}{2} |\vec{\Omega}(\omega)| \quad (19)$$

while its eigenvectors are¹

$$\begin{aligned} \hat{e}_+(\omega) &= \begin{pmatrix} \cos \theta(\omega) \\ e^{j\varphi(\omega)} \sin \theta(\omega) \end{pmatrix} \\ \hat{e}_-(\omega) &= \begin{pmatrix} \sin \theta(\omega) \\ -e^{j\varphi(\omega)} \cos \theta(\omega) \end{pmatrix} \end{aligned} \quad (20)$$

respectively, where $\theta(\omega)$ and $\varphi(\omega)$ are such that

$$\begin{aligned} \cos 2\theta(\omega) &= \frac{|q_1(\omega)|}{|\lambda_{\pm}(\omega)|} \\ \sin 2\theta(\omega) &= \frac{|q_2(\omega)|}{|\lambda_{\pm}(\omega)|} \\ \varphi(\omega) &= \arg\{q_2(\omega)\}. \end{aligned} \quad (21)$$

Given the unitary vector $\hat{e} = (e_x, e_y)^T$ in Jones space, the corresponding vector in Stokes space is $\vec{e} = (e_1, e_2, e_3)^T$ with

$$\begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = \begin{pmatrix} |e_x|^2 - |e_y|^2 \\ 2\Re\{e_x e_y^*\} \\ -2\Im\{e_x e_y^*\} \end{pmatrix}. \quad (22)$$

Therefore, taking into account (19) and (22), we find that the vectors corresponding to $\hat{e}_+(\omega)$ and $\hat{e}_-(\omega)$ in Stokes space are related to $\vec{\Omega}(\omega)$ according to

$$\begin{pmatrix} \Omega_1(\omega) \\ \Omega_2(\omega) \\ \Omega_3(\omega) \end{pmatrix} = \pm \begin{pmatrix} 2|q_1(\omega)| \\ 2\Re\{q_2(\omega)\} \\ 2\Im\{q_2(\omega)\} \end{pmatrix} \quad (23)$$

where the + sign refers to $\hat{e}_+(\omega)$. Thus, we can write the matrix $Q(\omega)$ in terms of the components of the PMD vector $\vec{\Omega}(\omega)$ as

$$Q(\omega) = \pm \frac{1}{2} \begin{pmatrix} \Omega_1(\omega) & \Omega_2(\omega) - j\Omega_3(\omega) \\ \Omega_2(\omega) + j\Omega_3(\omega) & -\Omega_1(\omega) \end{pmatrix} \quad (24)$$

where, by choosing the + sign, we assume that $\vec{\Omega}(\omega)$ is aligned with the fast PSP. We see that the uncertainty about the sign in (24) has no consequence as the sign of the components of $\vec{\Omega}(\omega)$ changes accordingly, leaving $Q(\omega)$ unchanged. Rewriting (16) as

$$\frac{dU(\omega)}{d\omega} = A(\omega)U(\omega) \quad (25)$$

where

$$A(\omega) \triangleq jQ(\omega) \quad (26)$$

we have the matrix differential equation to solve to find $U(\omega)$.

The matrix $A(\omega)$ in (26) depends upon the dispersion vector components $\Omega_i(\omega)$, $i = 1, 2, 3$, and, if we truncate to the m -th order its power series expansion about $\omega = 0$, we have

$$A(\omega) \simeq \sum_{k=0}^m \omega^k A_k \quad (27)$$

¹Note that these are two of the infinite eigenvectors of $Q(\omega)$. Indeed, any other vector pair of the form $e^{j\alpha} \hat{e}_{\pm}$ is its eigenvectors.

where the matrix coefficients $A_k = (1/k!)(d^k A(\omega)/d\omega^k)|_{\omega=0}$ depend on the parameters of the dispersion vector and its derivatives. Therefore, (27) corresponds to a $(m+1)$ -th order approximation on PMD and the solution of (25) provides the solution for any $(m+1)$ -th order PMD, whose statistics, however, are unfortunately currently unknown.

Second-order PMD is described by the derivative of $\vec{\Omega}(\omega)$ evaluated at $\omega = 0$ as follows [16]:

$$\vec{\Omega}_{\omega} \triangleq \left. \frac{d\vec{\Omega}(\omega)}{d\omega} \right|_{\omega=0} = \Delta\tau_{\omega} \vec{q} + \Delta\tau \vec{q}_{\omega} \quad (28)$$

where

$$\Delta\tau \triangleq |\vec{\Omega}(0)| \quad (29)$$

$$\Delta\tau_{\omega} \triangleq \left. \frac{d|\vec{\Omega}(\omega)|}{d\omega} \right|_{\omega=0} \quad (30)$$

$$\vec{q} \triangleq \frac{\vec{\Omega}(0)}{|\vec{\Omega}(0)|} \quad (31)$$

$$\vec{q}_{\omega} \triangleq \left. \frac{d}{d\omega} \frac{\vec{\Omega}(\omega)}{|\vec{\Omega}(\omega)|} \right|_{\omega=0}. \quad (32)$$

If we approximate $\vec{\Omega}(\omega)$ by its Taylor series expansion truncated to the first order (second-order approximation on PMD)

$$\vec{\Omega}(\omega) \simeq \vec{\Omega}_0 + \omega \vec{\Omega}_{\omega} \quad (33)$$

where $\vec{\Omega}_0 \triangleq \vec{\Omega}(0)$, and choose the coordinate reference system in such a way that $\vec{\Omega}_0$ is aligned along the positive direction of the s_1 axis in the Stokes space (fast PSP) and $\vec{\Omega}_{\omega}$ is parallel to the equatorial plane, we have

$$\vec{\Omega}(\omega) \simeq \begin{pmatrix} \Delta\tau + \Delta\tau_{\omega}\omega \\ \Delta\tau q_{\omega}\omega \\ 0 \end{pmatrix} \quad (34)$$

where $q_{\omega} \triangleq \pm |\vec{q}_{\omega}|$ is the rate of rotation of $\vec{\Omega}(\omega)$.

Due to the structure of $U(\omega)$ in (3), instead of solving (25), we can equivalently solve the following differential equation²:

$$\frac{d\hat{u}(\omega)}{d\omega} = A(\omega)\hat{u}(\omega) \quad \hat{u}(0) = \hat{u}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (35)$$

where

$$\hat{u}(\omega) \triangleq \begin{pmatrix} u_1(\omega) \\ -u_2^*(\omega) \end{pmatrix} \quad (36)$$

represents the first column of $U(\omega)$ and, for second-order PMD

$$\begin{aligned} A(\omega) &\simeq j \frac{1}{2} \begin{pmatrix} \Delta\tau + \Delta\tau_{\omega}\omega & \Delta\tau q_{\omega}\omega \\ \Delta\tau q_{\omega} & -\Delta\tau - \Delta\tau_{\omega}\omega \end{pmatrix} \\ &= A_0 + \omega A_1 \end{aligned} \quad (37)$$

$$A_0 \triangleq j \frac{\Delta\tau}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (38)$$

$$A_1 \triangleq j \frac{1}{2} \begin{pmatrix} \Delta\tau_{\omega} & \Delta\tau q_{\omega} \\ \Delta\tau q_{\omega} & -\Delta\tau_{\omega} \end{pmatrix}. \quad (39)$$

²The initial condition reflects the particular choice for $\vec{\Omega}(\omega)$ in (34), and the assumption that the carrier polarization state remains unchanged. In general, this is not true, and $U(\omega)$ should be pre- and postmultiplied by appropriate frequency-independent rotation matrices [10] to match laboratory measurements.

The solution of (35) gives an expression for the elements of the Jones matrix depending only on the parameters of the dispersion vector and its derivative, whose statistical properties are completely known [3]–[5].

III. SOLUTION OF THE DIFFERENTIAL EQUATION

Despite its seeming simplicity, a closed-form analytical solution to (35) is not known for $m \geq 1$, but if we seek a solution expandable in power series of the form

$$\hat{u}(\omega) = \hat{u}_0 + \omega \hat{u}_1 + \omega^2 \hat{u}_2 + \cdots + \omega^k \hat{u}_k + \cdots \quad (40)$$

we find that the initial condition is automatically satisfied and, by substituting (40) in (35) and equating the coefficients of the same power in ω , letting $\hat{u}_i = 0$ for $i < 0$, we obtain the recursive relationship for the series coefficients

$$\hat{u}_k = \frac{1}{k} \sum_{i=0}^{m-1} A_i \hat{u}_{k-1-i}, \quad \text{for } k = 1, 2, 3, \dots \quad (41)$$

from which we can recursively (and exactly) evaluate the vectors $\hat{u}_k$ to be used in (40). For any m , (41) turns out to be a numerically stable recursive relation, and the only problem in the numerical evaluation of (40) may show up for large values of ω when a large number of terms is required to achieve a prescribed accuracy and ω^k and $\hat{u}_k$ are at risk of overflowing and underflowing, respectively.

If we now take into consideration second-order PMD effects only, recursive relation (41) may be written as

$$\hat{u}_k = \frac{1}{k} (A_0 \hat{u}_{k-1} + A_1 \hat{u}_{k-2}), \quad \text{for } k = 1, 2, 3, \dots \quad (42)$$

which provides an *exact* solution for the second-order PMD case.

As a quick check, let us consider the case $\vec{\Omega}(\omega) = \text{constant}$ (first-order PMD), then $A_i = 0$ for $i \geq 1$, and from (41) or (42), we have

$$\hat{u}_k = \frac{1}{k!} A_0^k \hat{u}_0, \quad \text{for } k \geq 1$$

and the solution for first-order PMD becomes

$$\hat{u}(\omega) = \hat{u}_0 + \sum_{k=1}^{\infty} \frac{\omega^k}{k!} A_0^k \hat{u}_0 = e^{\omega A_0} \hat{u}_0. \quad (43)$$

As for a diagonal matrix $A = \text{diag}(\lambda_1, \dots, \lambda_n)$, it is $e^{\omega A} = \text{diag}(e^{\omega \lambda_1}, \dots, e^{\omega \lambda_n})$, and we have that

$$e^{\omega A_0} = \begin{pmatrix} e^{j\Delta\tau\omega/2} & 0 \\ 0 & e^{-j\Delta\tau\omega/2} \end{pmatrix}$$

and, thus

$$\hat{u}(\omega) = e^{\omega A_0} \hat{u}_0 = \begin{pmatrix} e^{j\Delta\tau\omega/2} \\ 0 \end{pmatrix}$$

and from (36), $u_1(\omega) = e^{j\omega\Delta\tau/2}$, $u_2(\omega) = 0$ which, substituted in (3), gives the known result

$$U(\omega) = \begin{pmatrix} e^{j\Delta\tau\omega/2} & 0 \\ 0 & e^{-j\Delta\tau\omega/2} \end{pmatrix}$$

as claimed.

A. Closed-Form Solution for a Particular Case

The previous recursive exact solution is susceptible of closed-form solution for a particular case, when the dispersion vector $\vec{\Omega}(\omega)$ does not rotate.

Indeed, consider again (25), for convenience rewritten here as

$$\frac{dD(\omega)}{d\omega} = A(\omega)D(\omega). \quad (44)$$

Letting $P(\omega) \triangleq \int_0^\omega A(\nu) d\nu$, if it were true that

$$\frac{d e^{P(\omega)}}{d\omega} = A(\omega) e^{P(\omega)} \quad (45)$$

then the solution to (44) would simply be

$$D(\omega) = e^{j \int_0^\omega Q(\nu) d\nu} D_0 \quad (46)$$

where $D_0 = D(0)$. Unfortunately, (45) is not always true when dealing with matrices. It turns out to be true only if $A(\omega)$ is of the form $A(\omega) = \omega^k A_0$ with A_0 a constant matrix, and not true if $A(\omega) = \sum_{k=k_1}^{k_2} \omega^k A_k$, unless the matrices A_k satisfy some particular relation. In the second-order case, $A(\omega)$ is as in (37), and then it can be easily demonstrated that (45) is true if the matrices A_0 and A_1 in (38) and (39), respectively, commute, i.e., if $A_0 A_1 = A_1 A_0$. It can be also easily seen that this is not the case, as $A_0 A_1 = (A_1 A_0)^T$, unless $q_\omega = 0$, i.e., if the dispersion vector $\vec{\Omega}(\omega)$ does not rotate, in which case $A_0 A_1$ turns out to be a diagonal matrix. Thus, only in this latter case the solution has a closed form, as given in (46), which, for $D_0 = I$ (i.e., under the assumption that the polarization state of the carrier remains unchanged), may be written as

$$D(\omega) = \begin{pmatrix} e^{j(\frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2)} & 0 \\ 0 & e^{-j(\frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2)} \end{pmatrix}. \quad (47)$$

Our result leads to the following considerations about some previously derived results.

In the literature [9], [10] the rotation of the dispersion vector $\vec{\Omega}(\omega)$ is taken into account through two complementary matrix rotations, $R(\omega)$ and $R^{-1}(\omega)$, and the fiber transfer matrix is normally written as

$$U(\omega) = R^{-1}(\omega) D(\omega) R(\omega) \quad (48)$$

where $D(\omega)$ is as in (47), and $R(\omega)$ is arbitrarily taken as [10]

$$R(\omega) = \begin{pmatrix} \cos(q_\omega\omega/4) & \sin(q_\omega\omega/4) \\ -\sin(q_\omega\omega/4) & \cos(q_\omega\omega/4) \end{pmatrix}. \quad (49)$$

In this case, the elements of the first row of the matrix U in (48) turn out to be expressed as

$$\begin{aligned} u_1(\omega) &= \cos \varphi(\omega) + j \cos \frac{1}{2} q_\omega \omega \sin \varphi(\omega) \\ u_2(\omega) &= j \sin \frac{1}{2} q_\omega \omega \sin \varphi(\omega) \end{aligned} \quad (50)$$

where $\varphi(\omega) = (1/2)\Delta\tau\omega + (1/4)\Delta\tau_\omega\omega^2$, and $D(\omega)$ in (47) and $R(\omega)$ in (49) may be written, after some algebra, as

$$D(\omega) = e^{j(\frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2)J} \quad (51)$$

$$R(\omega) = e^{\frac{1}{4}q_\omega\omega K} \quad (52)$$

where

$$\begin{aligned} J &\triangleq \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ K &\triangleq \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \end{aligned} \quad (53)$$

If (48) were correct,³ then, substituting (51) and (52) in (48), the resulting $U(\omega)$

$$U(\omega) = e^{-\frac{1}{4}q_\omega\omega K} e^{j(\frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2)J} e^{\frac{1}{4}q_\omega\omega K} \quad (54)$$

should satisfy (25), but we will see that this is not the case. That $U(\omega)$ in (48) is not correct could also be inferred from the fact that the effects of the rotation of the dispersion vector $\vec{\Omega}(\omega)$ cannot be disjointed from the differential delay $\Delta\tau$, as is evident from (37), and the fact that in (49), $\Delta\tau$ does not appear.

Let us now demonstrate that $U(\omega)$ in (48) does not satisfy (25). To this end we rewrite $U(\omega)$ as in (54) and use the following results. As it can be shown that

$$\begin{aligned} J^n K &= (-1)^n K J^n \\ K^n J &= (-1)^n J K^n \end{aligned} \quad (55)$$

it is simple to verify that for any scalar value x , the matrices J and K in (53) are such that

$$\begin{aligned} e^{xJ} K &= \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} J^n \right) K \\ &= K \left(\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!} J^n \right) = K e^{-xJ} \end{aligned} \quad (56)$$

and similarly that

$$e^{xK} J = J e^{-xK}. \quad (57)$$

Now, differentiating (54) with respect to ω , and using (56) and (57), we obtain, after some algebra

$$\begin{aligned} \frac{dU(\omega)}{d\omega} &= j\frac{1}{2}(\Delta\tau + \Delta\tau_\omega\omega) J e^{\frac{1}{2}q_\omega\omega K} U(\omega) \\ &\quad - \frac{1}{4}q_\omega K [U(\omega) - U^{-1}(\omega)] \end{aligned} \quad (58)$$

where we should have obtained $(dU(\omega)/d\omega) = A(\omega)U(\omega)$ with $A(\omega)$ as in (37). This proves that $U(\omega)$, in (48), is incorrect.

But, as $U(\omega)$ in (48) is widely used in the literature, let us have a closer look at (58). In the limit $q_\omega \rightarrow 0$, i.e., when the dispersion vector $\vec{\Omega}(\omega)$ rotates very slowly, as we can write

$$e^{\frac{1}{2}q_\omega\omega K} \simeq I + \frac{1}{2}q_\omega\omega K \quad (59)$$

then (58) becomes

$$\begin{aligned} \frac{dU(\omega)}{d\omega} &\simeq j\frac{1}{2} \left[(\Delta\tau + \Delta\tau_\omega\omega) J \right. \\ &\quad \left. + \frac{1}{2}(\Delta\tau q_\omega\omega + \Delta\tau_\omega q_\omega\omega^2) JK \right] U(\omega) \\ &\quad - \frac{1}{4}q_\omega K [U(\omega) - U^{-1}(\omega)] \end{aligned} \quad (60)$$

³Note that, as J and K do not commute, we cannot use the exponential properties valid in the scalar case, i.e., $e^{xJ} e^{yK} \neq e^{xJ+yK}$.

and, as $A(\omega)$ in (37) may also be written

$$A(\omega) = j\frac{1}{2}[(\Delta\tau + \Delta\tau_\omega\omega)J + \Delta\tau q_\omega\omega JK] \quad (61)$$

we see that the term which multiplies $U(\omega)$ in (60) is quite close to $A(\omega)$ for small values of ω (such that the ω^2 term is negligible) but there is a factor of two difference in the term multiplying JK , so it seems that one of the corrections proposed in [10] to the work in [9] really worsens the fiber transfer matrix approximation while improving the dispersion vector approximation. This may be explained by the fact that the relation between the two quantities is of differential nature. However, the last term in (60) should not be present and disappears only for $q_\omega = 0$. Then, we conclude that $U(\omega)$, as given in (48), gives approximate results which are reliable only when $\vec{\Omega}(\omega)$ rotates very slowly and only in a limited band around the carrier.

B. An Approximate Closed-Form Solution

We now derive an approximate closed-form solution to (25) for the second-order PMD.

As already said, the exact matrix $U(\omega)$ may be expressed as in (11), and then this is the form of the exact solution to (25). So letting

$$\begin{aligned} x(\omega) &= a(\omega)b_1(\omega) \\ y(\omega) &= a(\omega)b_2(\omega) \\ z(\omega) &= a(\omega)b_3(\omega) \end{aligned} \quad (62)$$

with⁴

$$a(\omega) = \text{sgn}\omega \sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)} \quad (63)$$

and substituting (62) and (63) in (11), we may write the elements $u_1(\omega)$ and $u_2(\omega)$ of the matrix $U(\omega)$ as

$$\begin{aligned} u_1(\omega) &= \cos \sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)} \\ &\quad + jx(\omega) \frac{\sin \sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}}{\sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}} \\ u_2(\omega) &= [z(\omega) + jy(\omega)] \\ &\quad \times \frac{\sin \sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}}{\sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}} \end{aligned} \quad (64)$$

and, thus, we change our problem from the determination of $u_1(\omega)$ and $u_2(\omega)$ to that of $x(\omega)$, $y(\omega)$ and $z(\omega)$. To this end, we take the derivative of (11), which, as $\exp[-ja(\omega)B(\omega)] = \cos a(\omega)I - j \sin a(\omega)B(\omega)$ and $B^2(\omega) = I$, may be written as

$$\begin{aligned} U'(\omega) &= j \left[a'(\omega)B(\omega) + \frac{1}{2} \sin 2a(\omega)B'(\omega) \right. \\ &\quad \left. - j \sin^2 a(\omega)B'(\omega)B(\omega) \right] U(\omega) \end{aligned} \quad (65)$$

⁴The choice of a particular sign for $a(\omega)$ is arbitrary as a rotation of an angle $2a$ about the vector $\vec{b}$ is equivalent to a rotation of an angle $-2a$ about $-\vec{b}$. The choice we make assures the continuity of $a(\omega)$ and $b_i(\omega)$, $i = 1, 2, 3$, and of their derivatives in $\omega = 0$.

where the prime indicates differentiation with respect to ω . By comparing (25) and (65), and taking into account (24) and (26), we have (omitting, for simplicity, the dependence on ω)

$$\begin{aligned} a'b_1 + \frac{1}{2}\sin(2a)b'_1 + \sin^2(a)(b'_2b_3 - b_2b'_3) &= \frac{1}{2}\Omega_1 \\ a'b_2 + \frac{1}{2}\sin(2a)b'_2 + \sin^2(a)(b_1b'_3 - b'_1b_3) &= \frac{1}{2}\Omega_2 \\ a'b_3 + \frac{1}{2}\sin(2a)b'_3 + \sin^2(a)(b'_1b_2 - b_1b'_2) &= \frac{1}{2}\Omega_3 \end{aligned} \quad (66)$$

and, by using (62) and (63), (66) becomes

$$\begin{aligned} x' &= f(a)(ax' - a'x) + g(a)(yz' - y'z) + \Omega_1/2 \\ y' &= f(a)(ay' - a'y) + g(a)(x'z - xz') + \Omega_2/2 \\ z' &= f(a)(az' - a'z) + g(a)(xy' - x'y) + \Omega_3/2 \end{aligned} \quad (67)$$

where

$$f(a) = \frac{a - \frac{1}{2}\sin(2a)}{a^2} \quad (68)$$

$$g(a) = \frac{\sin^2(a)}{a^2}. \quad (69)$$

Notice that (67) is a complex differential equation system more difficult to solve than (25), but, due to the structure of (64), an approximate solution as a power series expansion for $x(\omega)$, $y(\omega)$ and $z(\omega)$ would give better results than the corresponding solution for (25), given the same order of the expansion.

Indeed, as $a(0) = x(0) = y(0) = z(0) = 0$, from (67), we can recursively evaluate the n -th derivatives $x^{(n)}(0)$, $y^{(n)}(0)$, and $z^{(n)}(0)$ from the previous $n - 1$ derivatives. This leads to the following expansions

$$\begin{aligned} x(\omega) &= \frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2 + o(\omega^5) \\ y(\omega) &= \frac{1}{4}\Delta\tau q_\omega\omega^2 + o(\omega^4) \\ z(\omega) &= \frac{1}{24}\Delta\tau^2 q_\omega\omega^3 + o(\omega^5) \end{aligned} \quad (70)$$

where $o(\omega^k)$ is a term proportional to ω^k when $\omega \rightarrow 0$.

Retaining only the powers of order up to two, a first approximation for the functions $u_1(\omega)$ and $u_2(\omega)$ in (64) is

$$\begin{aligned} u_1(\omega) &\simeq \cos \sqrt{x^2(\omega) + y^2(\omega)} \\ &\quad + jx(\omega) \frac{\sin \sqrt{x^2(\omega) + y^2(\omega)}}{\sqrt{x^2(\omega) + y^2(\omega)}} \\ u_2(\omega) &\simeq jy(\omega) \frac{\sin \sqrt{x^2(\omega) + y^2(\omega)}}{\sqrt{x^2(\omega) + y^2(\omega)}} \end{aligned} \quad (71)$$

where

$$\begin{aligned} x(\omega) &= \frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2 \\ y(\omega) &= \frac{1}{4}\Delta\tau q_\omega\omega^2. \end{aligned} \quad (72)$$

As can be seen, this approximation gives $\Re\{u_2(\omega)\} = 0$, but a comparison with the numerical solution (40) shows an excellent agreement for $u_1(\omega)$ and $\Im\{u_2(\omega)\}$ for *any* range of parameters and angular frequency ω . We point out that (71) gives a very good approximation for $u_1(\omega)$ and $\Im\{u_2(\omega)\}$ as a whole; indeed, (72) are such that $\text{sgn}\omega\sqrt{x^2(\omega) + y^2(\omega)}$, or, rather, $\text{sgn}\omega \arccos(\cos \sqrt{x^2(\omega) + y^2(\omega)})$, approximates very well $a(\omega)$ for any angular frequency ω , while

$$\begin{aligned} b_1(\omega) &\simeq \text{sgn}(\omega) \frac{x(\omega)}{\sqrt{x^2(\omega) + y^2(\omega)}} \\ b_2(\omega) &\simeq \text{sgn}(\omega) \frac{y(\omega)}{\sqrt{x^2(\omega) + y^2(\omega)}} \end{aligned} \quad (73)$$

are well approximated in a bandwidth of the order of $1/\Delta\tau$, and outside this bandwidth, (73) only gives an approximation for the *envelope* of the real $b_1(\omega)$ and $b_2(\omega)$. However, when $x(\omega)$ and $y(\omega)$ in (72) are used in (71), they give an approximation for $u_1(\omega)$ and $\Im\{u_2(\omega)\}$, which turns out to be excellent for any angular frequency ω . This means that $z^2(\omega)$ is negligible with respect to $x^2(\omega) + y^2(\omega)$ for almost all values of ω . We conclude that, when ω increases, an approximation for $z(\omega)$ should follow the same behavior of the approximations in (72), and must turn into a function going up, at most, to ω^2 . As numerical evidence shows that $x(\omega)$, $y(\omega)$, and $a(\omega)$ are already well suited, we tried to determine a better approximation than that in (70) for $z(\omega)$, possibly valid over a broader range of frequencies.

After many failed attempts at an analytical insight from (67), we postulated the following approximation⁵ for $z(\omega)$

$$z(\omega) \simeq \frac{1}{8}\Delta\tau q_\omega\omega^2 \text{erf}\left(\frac{\sqrt{\pi}}{6}\Delta\tau\omega\right) \quad (74)$$

where the error function was chosen among various alternatives because, in this way, the first four derivatives of (74) coincide with that of $z(\omega)$ in (70), and for $\omega \rightarrow \infty$, it shows the desired parabolic increase.

Our final approximation is, then, a simple adjustment of (71) as

$$\begin{aligned} u_1(\omega) &\simeq \cos \sqrt{x^2(\omega) + y^2(\omega)} \\ &\quad + jx(\omega) \frac{\sin \sqrt{x^2(\omega) + y^2(\omega)}}{\sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}} \\ u_2(\omega) &\simeq [z(\omega) + jy(\omega)] \frac{\sin \sqrt{x^2(\omega) + y^2(\omega)}}{\sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}} \end{aligned} \quad (75)$$

where

$$\begin{aligned} x(\omega) &= \frac{1}{2}\Delta\tau\omega + \frac{1}{4}\Delta\tau_\omega\omega^2 \\ y(\omega) &= \frac{1}{4}\Delta\tau q_\omega\omega^2 \\ z(\omega) &= \frac{1}{8}\Delta\tau q_\omega\omega^2 \text{erf}\left(\frac{\sqrt{\pi}}{6}\Delta\tau\omega\right) \end{aligned} \quad (76)$$

⁵In the expansions (70), for any order, q_ω always appears multiplied by $\Delta\tau$, and we take this into account in the approximation.

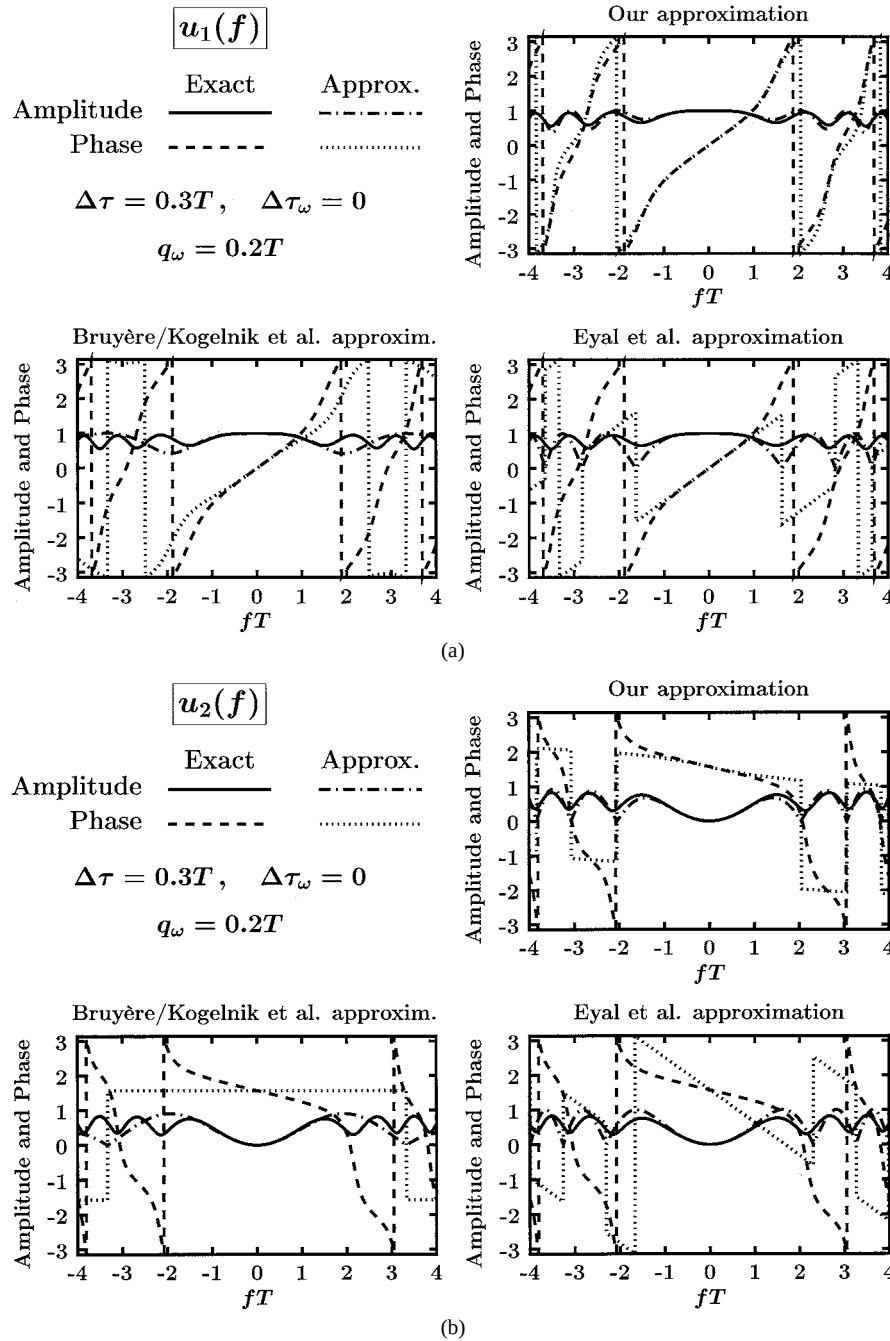


Fig. 1 Exact solution versus approximations.

and we have inserted $z^2(\omega)$ in the square root at the denominator only to preserve the unity of the matrix $U(\omega)$.

It is interesting to compare our approximate solution with that in [12], which gives $U(\omega)$ as an exponential expansion that, to the second order, is

$$U(\omega) = U_0 e^{\omega N_1} e^{\frac{1}{2}\omega^2 N_2} \quad (77)$$

where

$$U_0 = U(0) \quad N_1 = U_0^{-1} \left. \frac{dU(\omega)}{d\omega} \right|_{\omega=0}$$

$$N_2 = U_0^{-1} \left. \frac{d^2 U(\omega)}{d\omega^2} \right|_{\omega=0} - N_1^2.$$

Notice that, if N_1 and N_2 had commuted (i.e., if it were possible to write $e^{\omega N_1} e^{(1/2)\omega^2 N_2} = e^{\omega N_1 + (1/2)\omega^2 N_2}$), then the elements $u_1(\omega)$ and $u_2(\omega)$ of the matrix $U(\omega)$ obtained in this way would have been the same as in (71). Because this not the case, they instead turn out to be

$$u_1(\omega) = \frac{1}{2} \left(1 + \frac{\Delta\tau_\omega}{D} \right) e^{j(\Delta\tau\omega/2 + D\omega^2/4)} + \frac{1}{2} \left(1 - \frac{\Delta\tau_\omega}{D} \right) e^{j(\Delta\tau\omega/2 - D\omega^2/4)}$$

$$u_2(\omega) = \frac{1}{2} \frac{\Delta\tau q_\omega}{D} \left(e^{-j(\Delta\tau\omega/2 - D\omega^2/4)} - e^{-j(\Delta\tau\omega/2 + D\omega^2/4)} \right) \quad (78)$$

where

$$D = \sqrt{\Delta\tau_\omega^2 + \Delta\tau^2 q_\omega^2}. \quad (79)$$

We will see, in the following, that approximation (78) is superior to that given by (50), even if slightly worse than our approximation (75), especially for the term $u_2(\omega)$, i.e., the term taking into account the coupling between the orthogonal polarizations.

IV. COMPARISON BETWEEN EXACT SOLUTION AND VARIOUS APPROXIMATIONS

We performed extensive checks over very broad ranges for the parameters $\Delta\tau$, $\Delta\tau_\omega$, and q_ω , always obtaining the same degree of approximation over a wide range of angular frequencies ω , so we present here only a representative selection of the results we obtained. Because the Jones transfer matrix $U(\omega)$ in (3) is completely determined by its elements $u_1(\omega)$ and $u_2(\omega)$, we compare their exact amplitude and phase as obtained numerically from the expansion (40) with those obtained from the approximations (50), (75), and (78). In the examples shown, all parameters, including the frequency $f = (\omega/2\pi)$, are normalized to the arbitrary time T , which may be regarded as the bit time of an optical communication system.

Because all three approximations—ours in (75) and those given by (50) and (78)—agree when the dispersion vector does not rotate, i.e., when $q_\omega = 0$, we begin by comparing them for $\Delta\tau_\omega = 0$ and $q_\omega \neq 0$. For second-order PMD, in Fig. 1(a) and (b), the exact amplitude and phase of $u_1(\omega)$ and $u_2(\omega)$, respectively, are shown together with the aforesaid approximations for $\Delta\tau = 0.3T$, $\Delta\tau_\omega = 0$, and $q_\omega = 0.2T$. As can be seen, approximation (75) is the most precise among all three, and the approximation goodness is maintained over a wider bandwidth, whereas the other two are acceptable in a frequency bandwidth of the order of $1/T$.

In Fig. 2, the unwrapped phases of $u_1(\omega)$ and $u_2(\omega)$ of Fig. 1(a) and (b) are shown together with some further approximations which may be derived from (75) as follows. From (75), the approximate phase of the element $u_1(\omega)$ is

$$\varphi_1(\omega) = \arctan \left(\frac{x(\omega)}{\sqrt{x^2(\omega) + y^2(\omega) + z^2(\omega)}} \times \tan \sqrt{x^2(\omega) + y^2(\omega)} \right) \quad (80)$$

and, because for small frequencies, $x(\omega)$ is dominant over the other terms, it may be approximated as

$$\begin{aligned} \varphi_1(\omega) &\simeq \varphi_{1a}(\omega) \\ &= \arctan \left(\text{sgn}[x(\omega)] \tan \sqrt{x^2(\omega) + y^2(\omega)} \right) \\ &= \text{sgn}[x(\omega)] \frac{1}{2} \Delta\tau |\omega| \\ &\quad \times \sqrt{1 + \frac{\Delta\tau_\omega}{\Delta\tau} \omega + \frac{1}{4} \left(\frac{\Delta\tau_\omega^2}{\Delta\tau^2} + q_\omega^2 \right) \omega^2} \end{aligned} \quad (81)$$

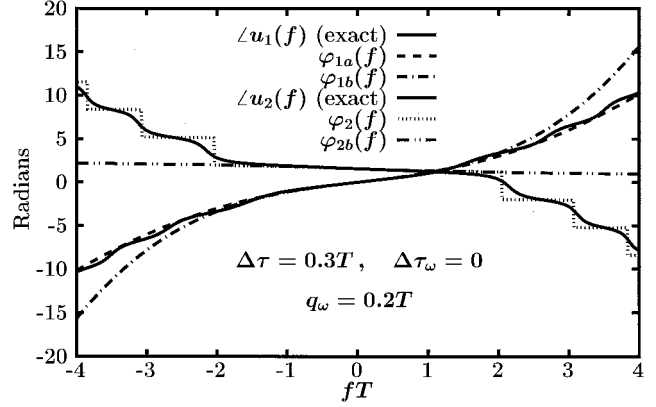


Fig. 2. Unwrapped phases of u_1 and u_2 in Fig. 1(a) and (b).

which, in a narrower bandwidth around the carrier, may be further approximated as

$$\begin{aligned} \varphi_{1a}(\omega) &\simeq \varphi_{1b}(\omega) \\ &= \text{sgn}[\omega x(\omega)] \left[\frac{1}{2} \Delta\tau \omega + \frac{1}{4} \Delta\tau_\omega \omega^2 \right. \\ &\quad \left. + \frac{1}{16} \left(\frac{\Delta\tau_\omega^2}{\Delta\tau} + \Delta\tau q_\omega \right) \omega^3 \right]. \end{aligned} \quad (82)$$

The phase of $u_2(\omega)$ turns out to be

$$\varphi_2(\omega) = \arctan \frac{y(\omega)}{z(\omega)} = \frac{\pi}{2} - \arctan \frac{z(\omega)}{y(\omega)} \quad (83)$$

and, as $z(\omega)$ is almost always smaller than $y(\omega)$, it may be approximated as

$$\varphi_2(\omega) \simeq \varphi_{2a}(\omega) = \frac{\pi}{2} - \frac{z(\omega)}{y(\omega)} \quad (84)$$

which, in a narrow bandwidth around the carrier, may be further approximated as

$$\varphi_{2a}(\omega) \simeq \varphi_{2b}(\omega) = \frac{\pi}{2} - \frac{1}{6} \Delta\tau \omega. \quad (85)$$

It may be seen that the phase of $u_2(\omega)$ is mostly linear.

In Fig. 2, $\varphi_{1a}(\omega)$ in (80) is not reported as preventing cluttering in the drawing, but it is almost coincident with the real phase of $u_1(\omega)$, as can be seen from Fig. 1(a). From these figures, we see that, when $\Delta\tau_\omega = 0$, the effect of PMD is mainly a phase distortion of the signal in one polarization to which a time shifted replica of the second derivative⁶ of the signal in the other polarization is summed. Moreover, the phase distortion is more like of the f^3 type, rather than of f^2 type as, for example, is the distortion due to the fiber chromatic dispersion.

Let us now see what happens when the dispersion vector rate of rotation increases. Fig. 3(a) and (b) refers to the case $\Delta\tau = 0.3T$, $\Delta\tau_\omega = 0$ and $q_\omega = 0.8T$, and it can be seen that while (75) still approximates very well the amplitude and

⁶This, of course, will depend on the signal bandwidth.

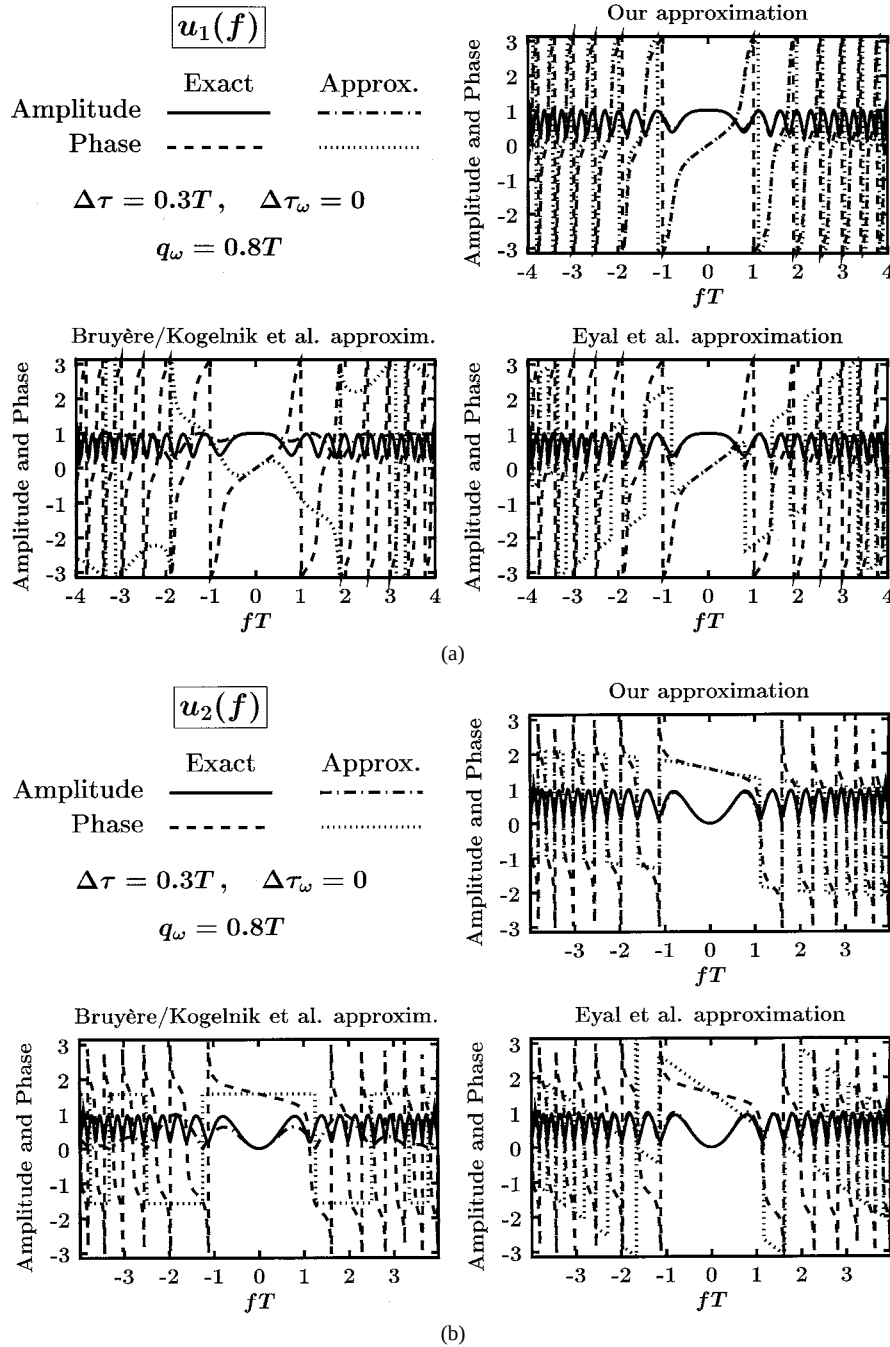
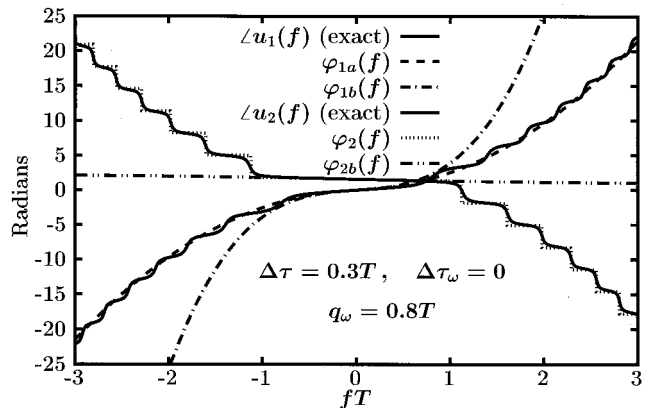


Fig. 3. Exact solution versus approximations.

phase of $u_1(\omega)$ and $u_2(\omega)$ for any frequency, (50) is acceptable only over a bandwidth of the order of $(1/2T)$, whereas (78) approximates very well the amplitudes but somehow misses the phases. We show in Fig. 4 the unwrapped phases of $u_1(\omega)$ and $u_2(\omega)$ in Fig. 3(a) and (b), along with the approximations (81)–(85), omitting again (80), practically coincident with the real phase of $u_1(\omega)$. As in Fig. 2, the phase of $u_1(\omega)$ goes as $\text{sgn}(f)f^2$ and, thus, its effects cannot be regarded as those of a chromatic dispersion for $\Delta\tau_\omega = 0$ (which is the most probable value [4]).

Things change somewhat when $\Delta\tau_\omega \neq 0$. In Fig. 5(a) and (b), the amplitude and phase of $u_1(\omega)$ and $u_2(\omega)$ for the case $\Delta\tau = 0.3T, \Delta\tau_\omega = 0.1T^2$, and $q_\omega = 0.8T$ are reported. Al-

Fig. 4. Unwrapped phases of u_1 and u_2 in Fig. 3(a) and (b).

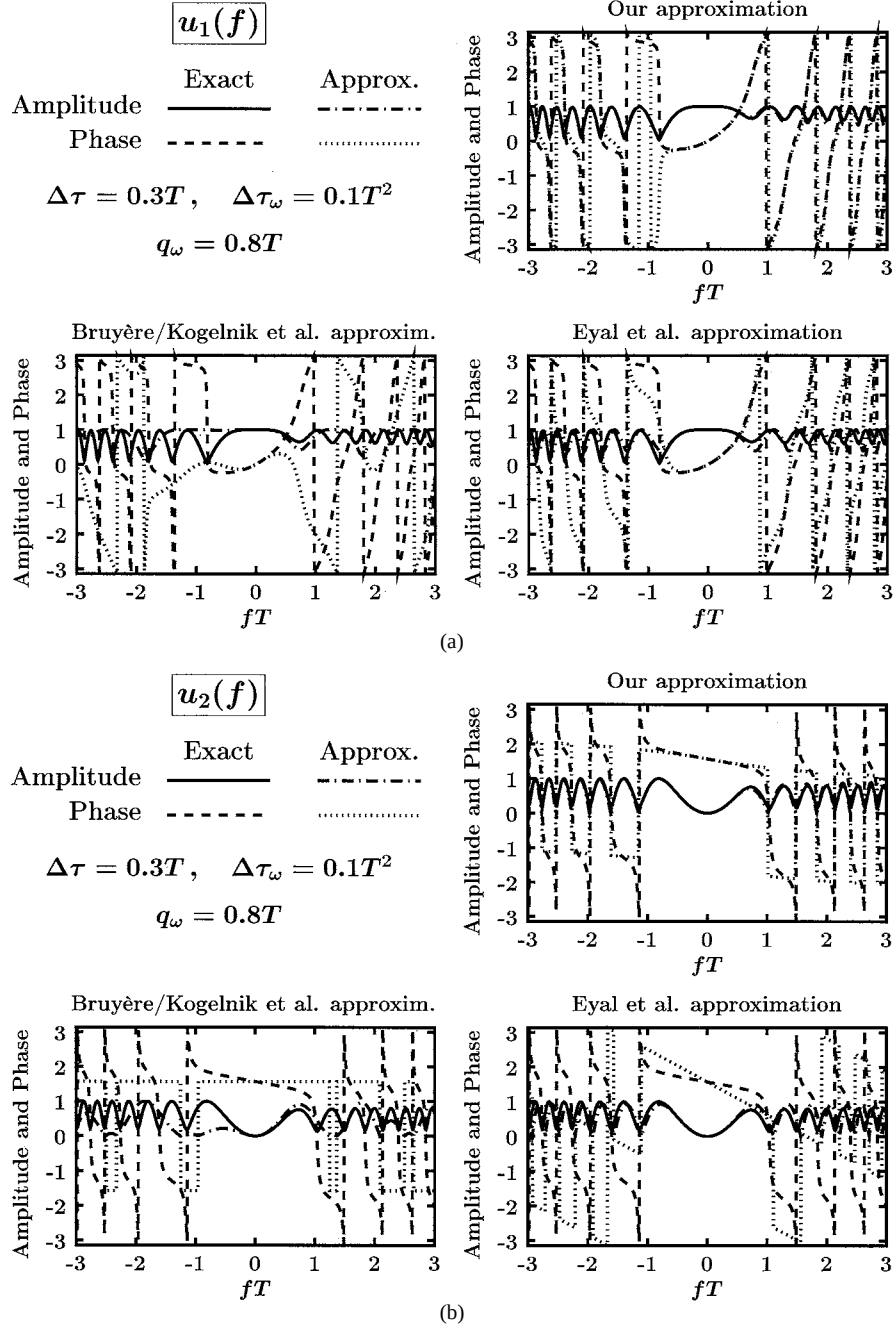
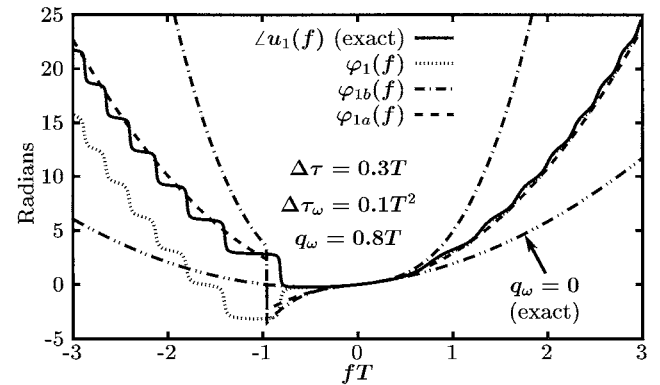


Fig. 5 Exact solution versus approximations.

though what has already been said about the accuracy of the approximations is maintained, it can be seen that the kind of filtering performed by $u_1(\omega)$ and $u_2(\omega)$ is not symmetric any longer, and frequencies above or below the carrier are treated differently, both in amplitude and phase. This kind of asymmetry is evident from Fig. 6, which reports the unwrapped phase of $u_1(\omega)$ together with approximations (80)–(82), as well as the phase of $u_1(\omega)$ for the same parameters but with $q_\omega = 0$. Regarding (80), there is a seeming discrepancy with the real phase of $u_1(\omega)$ which, for decreasing values of frequency starting from zero, exhibits a first jump of π radians, while, correspondingly, (80) shows a jump of $-\pi$ radians and then follows the behavior of the real phase with a difference of exactly 2π radians.

Fig. 6. Unwrapped phases of u_1 in Fig. 5(a).

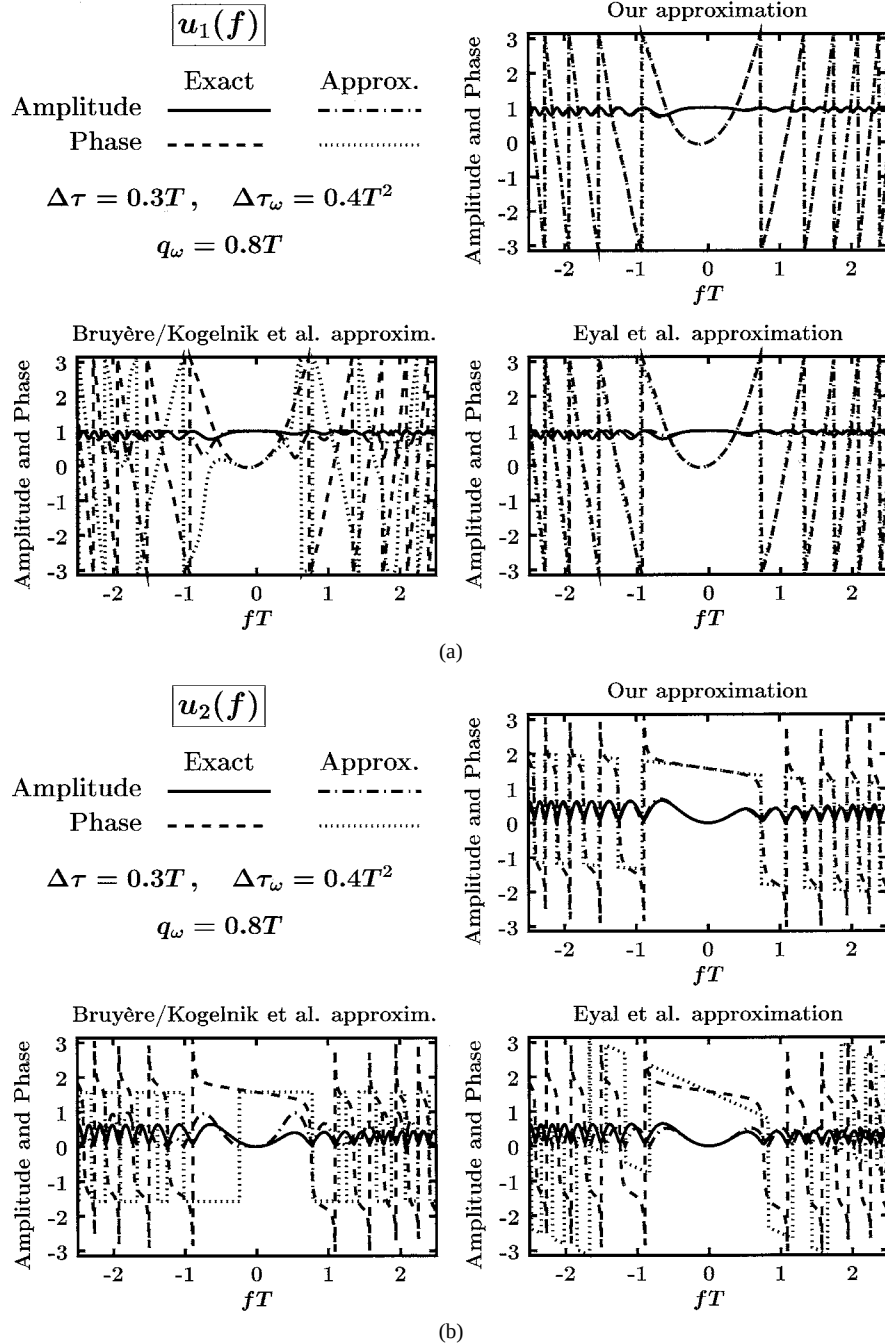


Fig. 7. Exact solution versus approximations.

So, apart from the different direction of the first jump, the two phases may be regarded as coincident. As can be seen, the real phase of $u_1(\omega)$ can be roughly approximated by a parabola and by comparing it with the phase for $q_\omega = 0$ (not rotating dispersion vector Ω), which is of parabolic shape. It seems that the rotation of Ω enhances the PMD-induced chromatic fiber dispersion, but one should also consider that, for $f < 0$, apart from the jumps, the phase remains nearly constant and, depending on the signal bandwidth, it could not be seen as parabolic.

Increasing the value of $\Delta\tau_\omega$, we obtain the results shown in Fig. 7(a) and (b) for $\Delta\tau = 0.3T$, $\Delta\tau_\omega = 0.4T^2$ and $q_\omega = 0.8T$. In this case, approximation (75) remains excellent and

(50) is acceptable only in a very narrow bandwidth, whereas (78) is of almost the same quality of (75) in terms of $u_1(\omega)$, but poorer than (75) in terms of $u_2(\omega)$. From Fig. 8, reporting the unwrapped phase of $u_1(\omega)$ and $u_2(\omega)$ in Fig. 7(a) and (b) together with approximations (80)–(83) and the phase of $u_1(\omega)$ for the same parameters but with $q_\omega = 0$, we can see that when $\Delta\tau_\omega$ increases, the rotation of Ω has nearly no effects on the signal propagating in one polarization, its only effect being a coupling between the two polarizations.

From numerical results, we see that (75) is, in general, an excellent approximation, and more accurate than those previously proposed.

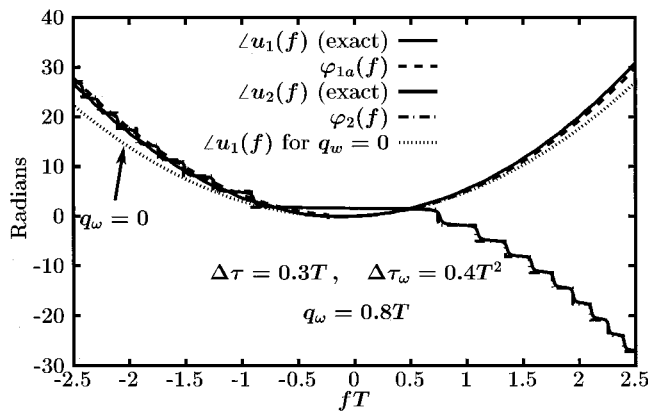


Fig. 8. Unwrapped phases of u_1 and u_2 in Fig. 7(a) and (b).

V. CONCLUSION

PMD will become increasingly important in determining the system performance as the bit rate and distance increase; thus, a deeper understanding of its effects is necessary. In this paper, we present a technique that permits the evaluation of the Jones transfer matrix of a fiber, taking into account the effects of arbitrary-order PMD exactly. Due to its stochastic nature, this result is somehow impaired without knowledge of the statistics of the parameters governing the phenomenon. These statistics are currently known only for second-order PMD, and for this case, we develop an accurate closed-form analytical approximation that can be effectively used to state the system performance analytically or by simulation. Until now, the only way to state PMD effects is through simulation, where either the effects of the various orders of PMD are taken into account as a whole or approximate solutions for the second-order PMD are incorporated into simulation. The tool we develop could help improve understanding of how important the effects of PMD up to the second order are, either by more accurate simulation or by extending the analytical approach presented in [13] to evaluate the bit-error probability of the system. In this paper, we show that second-order PMD causes primarily a phase distortion that cannot always be regarded as a chromatic dispersion; indeed, when $\Delta\tau_w \simeq 0$, it is more like a f^3 phase distortion. The dispersion vector rotation enhances the effects of a small $\Delta\tau_w$ (responsible for the PMD-induced chromatic fiber dispersion) and, in this case, the distortion is *almost* like a chromatic dispersion. However, when $\Delta\tau_w$ is large, the dispersion vector rotation is nearly negligible and its effects are primarily a coupling between the two orthogonal polarizations. It remains to be investigated what the dominant effect may be.

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Enrico Forestieri (S'91–M'92) was born in Milazzo, Italy, in 1960. He received the Dr. Ing. degree in electronics engineering from the University of Pisa, Pisa, Italy, in 1988.

From 1989 to 1991, he was a Postdoctoral Scholar at the University of Parma, Parma, Italy, working in optical communication systems. From 1991 to 2000, he was a Research Scientist and faculty member at the University of Parma. He is now Associate Professor of Telecommunications at the Scuola Superiore Sant'Anna, Pisa, Italy. His

research interests are in the area of digital communication theory and optical communication systems.



Luca Vincetti was born in Parma, Italy, in 1969. He received the degree in electronic engineering from the University of Parma in 1996. He received the Ph.D. degree in information technology from the Dipartimento di Ingegneria dell'Informazione (DII), University of Parma, in January 2000.

Since February 2000, he has been a researcher at the Consorzio Nazionale Interuniversitario per le Telecomunicazioni (CNIT), Pisa, Italy. His current research includes nonlinear optics, integrated optics, erbium-doped devices, polarization mode dispersion, and finite-element propagation analysis of electromagnetic fields.